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United States Patent	12403995
Kind Code	B2
Date of Patent	September 02, 2025
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Outboard motor

Abstract

An outboard motor includes an engine, an upper case, a lower case, a cavitation plate, a transmission, a cooling water passage, a first pump, and a second pump. The transmission includes a drive shaft and a propeller shaft. The cooling water passage is connected to the engine and located in the upper case and the lower case. The first pump is above the cavitation plate. The first pump is connected to the drive shaft and operable to send cooling water to the engine through the cooling water passage. The second pump is drivable by the driving force from the transmission to send the cooling water to the engine through the cooling water passage.

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Appl. No.: 17/979035

Filed: November 02, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230312073 A1	Oct. 05, 2023

Foreign Application Priority Data

JP	2022-062042	Apr. 01, 2022
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Publication Classification

Int. Cl.: B63H20/14 (20060101); B63H20/02 (20060101); B63H20/30 (20060101); B63H20/32 (20060101)

U.S. Cl.:

CPC B63H20/14 (20130101); B63H20/02 (20130101); B63H20/30 (20130101); B63H20/32 (20130101);

Field of Classification Search

CPC: B63H (20/14); B63H (20/02); B63H (20/30); B63H (20/32)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2022-062042, filed on Apr. 1, 2022. The contents of that application are incorporated by reference herein in their entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present invention relates to an outboard motor.

2. Description of the Related Art

(3) An outboard motor includes a water pump and a cooling water passage for supplying cooling water to the engine. A water intake is provided at the bottom of the outboard motor, and the cooling water passage is connected to the engine and the water intake. The water pump delivers the cooling water to the engine through the cooling water passage.

(4) On the other hand, as disclosed in JP-A-2015-67191, some outboard motors include a main pump and a sub-pump. The sub-pump sends cooling water to the main pump, and the main pump sends the cooling water to the engine. In the outboard motor disclosed in JP-A-2015-67191, the main pump is arranged in the upper case. The main pump is driven by an electric motor. The sub-pump is arranged in the lower case. The sub-pump is driven by a rotation of the drive shaft.

SUMMARY OF THE INVENTION

(5) In a case where the main pump is driven by an electric motor as in the above outboard motor, a space is required in the outboard motor to accommodate the electric motor. As a result, the size of the outboard motor is increased.

(6) Preferred embodiments of the present invention efficiently send cooling water to engines of outboard motors by a plurality of pumps while reducing the size of the outboard motors.

(7) An outboard motor according to a preferred embodiment of the present invention includes an engine, an upper case, a lower case, a cavitation plate, a transmission, a cooling water passage, a first pump, and a second pump. The upper case is below the engine. The lower case is below the upper case. The cavitation plate is connected to the lower case. The transmission includes a drive shaft and a propeller shaft. The drive shaft extends downward from the engine and is located in the upper case and the lower case. The propeller shaft is connected to the drive shaft, extends in a front-rear direction of the outboard motor, and is located in the lower case. The cooling water passage is connected to the engine and located in the upper case and the lower case. The first pump is above the cavitation plate. The first pump is connected to the drive shaft and is operable to send cooling water to the engine through the cooling water passage. The second pump is drivable by a driving force from the transmission to send the cooling water to the engine through the cooling water passage.

(8) In an outboard motor according to a preferred embodiment of the present invention, the cooling water is efficiently sent to the engine by the first pump and the second pump. Also, the first pump is connected to the drive shaft and driven by the rotation of the drive shaft. Therefore, the outboard motor is smaller than when the first pump is driven by an electric motor.

(9) The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a side view of an outboard motor according to a first preferred embodiment of the present invention.

(2) FIG. 2 is a block diagram showing a cooling system of the outboard motor.

(3) FIG. 3 is a side view of an outboard motor according to a second preferred embodiment of the present invention.

(4) FIG. 4 is a side view of an outboard motor according to a third preferred embodiment of the present invention.

(5) FIG. 5 is a side view of an outboard motor according to a fourth preferred embodiment of the present invention.

(6) FIG. 6 is a side view of an outboard motor according to a fifth preferred embodiment of the present invention.

(7) FIG. 7 is a side view of an outboard motor according to a sixth preferred embodiment of the present invention.

(8) FIG. 8 is a side view of an outboard motor according to a seventh preferred embodiment of the present invention.

(9) FIG. 9 is a side view of an outboard motor according to an eighth preferred embodiment of the present invention.

(10) FIG. 10 is a side view of an outboard motor according to a ninth preferred embodiment of the present invention.

(11) FIG. 11 is a side view of an outboard motor according to a tenth preferred embodiment of the

present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(12) Preferred embodiments will be described below with reference to the drawings. FIG. 1 is a side view of an outboard motor **1** according to a first preferred embodiment. The outboard motor **1** includes a bracket **2**, an engine **3**, a transmission mechanism **4**, an engine cowl **5**, an upper case **6**, and a lower case **7**. The outboard motor **1** is attached to a watercraft via the bracket **2**.

(13) The engine **3** generates thrust to propel the watercraft. The engine **3** includes a crankshaft **11**. The crankshaft **11** extends in the vertical direction of the outboard motor **1**. The transmission mechanism **4** transmits the driving force of the engine **3** to a propeller **12**. The transmission mechanism **4** includes a drive shaft **8**, a propeller shaft **9**, and a shift mechanism **10**. The drive shaft **8** is connected to the crankshaft **11**. The drive shaft **8** extends in the vertical direction of the outboard motor **1**. The drive shaft **8** extends downward from the engine **3**.

(14) The propeller shaft **9** extends in a front-rear direction of the outboard motor **1**. The propeller shaft **9** is connected to the drive shaft **8** via the shift mechanism **10**. The propeller **12** is attached to the propeller shaft **9**. The shift mechanism **10** includes a plurality of gears and a clutch. The shift mechanism **10** switches the transmission direction of the rotation from the drive shaft **8** to the propeller shaft **9**. As a result, the watercraft is switched between forward and reverse.

(15) The engine **3** is disposed in the engine cowl **5**. The upper case **6** is disposed below the engine **3**. The lower case **7** is disposed below the upper case **6**. The drive shaft **8** is disposed in the upper case **6** and the lower case **7**. The propeller shaft **9** and the shift mechanism **10** are disposed in the lower case **7**. Specifically, the lower case **7** includes a torpedo portion **13**. The torpedo portion **13** has an outwardly bulging shape. The propeller shaft **9** and the shift mechanism **10** are disposed in the torpedo portion **13**. The cavitation plate **14** is connected to the lower case **7**. The cavitation plate **14** protrudes rearward from the lower case **7**.

(16) FIG. 2 is a block diagram showing a cooling system of the outboard motor **1**. As shown in FIG. 2, the engine **3** includes a water jacket **15**. The cooling water flowing through the water jacket **15** cools the engine **3**. The outboard motor **1** includes a cooling water passage **16** and a drain passage **17**. The cooling water passage **16** and the drain passage **17** are connected to the water jacket **15** of the engine **3**. The cooling water passage **16** and the drain passage **17** are disposed in the upper case **6** and the lower case **7**.

(17) The cooling water is supplied from the outside of the outboard motor **1** to the water jacket **15** of the engine **3** through the cooling water passage **16**. The cooling water is discharged from the water jacket **15** to the outside of the outboard motor **1** through the drain passage **17** and the drain port **18**. The drain port **18** is provided in the lower case **7**, for example. Alternatively, the drain port **18** may be provided in the upper case **6**.

(18) The cooling water passage **16** includes a water intake **21**, a first passage **22**, and a second passage **23**. The outboard motor **1** includes a first pump **25**, a second pump **26**, and a valve **27**. The water intake **21** is provided in the lower case **7**. The water intake **21** communicates with the outside of the outboard motor **1**. The cooling water is drawn into the cooling water passage **16** from the outside of the outboard motor **1** through the water intake **21**.

(19) The first passage **22** is connected to the water intake **21**. The first pump **25** is provided in the first passage **22**. The second passage **23** is connected to the water intake **21**. The second pump **26** is provided in the second passage **23**. The valve **27** is provided in the second passage **23**. The valve **27** allows the cooling water to flow from the second passage **23** to the first passage **22**. The valve **27** inhibits the flow of the cooling water from the first passage **22** to the second passage **23**.

(20) The first passage **22** and the second passage **23** are connected to the water jacket **15**. The first pump **25** sends the cooling water to the engine **3** through the first passage **22**. The second pump **26** sends the cooling water to the engine **3** through the second passage **23**.

(21) As shown in FIG. 1, the first pump **25** is coaxial with the drive shaft **8**. The first pump **25** is connected to the drive shaft **8**. The first pump **25** is driven by the rotation of the drive shaft **8**. The

first pump 25 is disposed below the lower end of the bracket 2. The first pump 25 is disposed in the lower case 7. The first pump 25 is disposed above the cavitation plate 14. The first pump 25 is disposed between the upper end 90 of the lower case 7 and the cavitation plate 14. The first pump 25 is disposed above the waterline L1 where the watercraft is stopped.

(22) The second pump 26 is disposed in the lower case 7. The second pump 26 is disposed below the first pump 25. The second pump 26 is disposed below the cavitation plate 14. The second pump 26 is disposed below the waterline L1 where the watercraft is stopped. The second pump 26 is disposed in the torpedo portion 13.

(23) The second pump 26 is connected to the propeller shaft 9. The second pump 26 is coaxial with the propeller shaft 9. The second pump 26 is driven by the rotation of the propeller shaft 9. The first pump 25 is a positive displacement pump such as a gear pump or a vane pump. The second pump 26 is a non-positive displacement pump such as a centrifugal pump or an axial pump.

(24) In the outboard motor 1 according to the present preferred embodiment, when the engine 3 is started, the first pump 25 is driven by the rotation of the drive shaft 8. The first pump 25 sends the cooling water from the water intake 21 to the water jacket 15 of the engine 3 through the first passage 22. Further, the rotation of the propeller shaft 9 drives the second pump 26. Thus, the second pump 26 sends the cooling water from the water intake 21 to the water jacket 15 of the engine 3 through the second passage 23.

(25) In the outboard motor 1 according to the present preferred embodiment, the cooling water is sent to the engine 3 by the first pump 25 and the second pump 26. Thus, the cooling water is sent to the engine 3 efficiently. Also, the first pump 25 is connected to the drive shaft 8 and driven by the rotation of the drive shaft 8. Therefore, the size of the outboard motor 1 is reduced compared to when an electric motor for driving the first pump 25 is provided.

(26) Although a preferred embodiment of the present invention has been described above, the present invention is not limited to the above preferred embodiment, and various modifications are possible without departing from the gist of the present invention.

(27) The arrangement of the first pump 25 and the second pump 26 is not limited to that of the first preferred embodiment, and may be changed. For example, the second pump 26 may be disposed above the cavitation plate 14. FIG. 3 is a side view of the outboard motor 1 according to a second preferred embodiment. As shown in FIG. 3, in the outboard motor 1 according to the second preferred embodiment, the second pump 26 is connected to the crankshaft 11. The second pump 26 is disposed in the engine cowl 5. Alternatively, the second pump 26 may be disposed in the upper case 6.

(28) FIG. 4 is a side view of the outboard motor 1 according to a third preferred embodiment. As shown in FIG. 4, in the outboard motor 1 according to the third preferred embodiment, the transmission mechanism 4 includes an intermediate shaft 31 and a link mechanism 32. The intermediate shaft 31 is offset or eccentrically mounted with respect to the drive shaft 8. The intermediate shaft 31 is connected to the drive shaft 8 via the link mechanism 32 such as gears or a belt. The second pump 26 is coaxial with the intermediate shaft 31. The second pump 26 is connected to the intermediate shaft 31. The second pump 26 is disposed above the cavitation plate 14 in the lower case 7.

(29) FIG. 5 is a side view of the outboard motor 1 according to a fourth preferred embodiment. The second pump 26 may be connected to the drive shaft 8, as shown in FIG. 5. In the outboard motor 1 according to the fourth preferred embodiment, the second pump 26 is coaxial with the drive shaft 8. The second pump 26 is disposed above the cavitation plate 14. The second pump 26 is disposed above the first pump 25.

(30) FIG. 6 is a side view of the outboard motor 1 according to a fifth preferred embodiment. As shown in FIG. 6, in the outboard motor 1 according to the fifth preferred embodiment, the second pump 26 is disposed below the cavitation plate 14. The second pump 26 is coaxial with the drive shaft 8. The second pump 26 is connected to the drive shaft 8.

(31) The configuration of the transmission mechanism **4** is not limited to that of the above preferred embodiments, and may be modified. For example, FIG. **7** is a side view of the outboard motor **1** according to a sixth preferred embodiment. As shown in FIG. **7**, in the outboard motor **1** according to the sixth preferred embodiment, the drive shaft **8** includes a first drive shaft **8A** and a second drive shaft **8B**. The first drive shaft **8A** is connected to the crankshaft **11**. The second drive shaft **8B** is connected to the propeller shaft **9** via the shift mechanism **10**. The first drive shaft **8A** and the second drive shaft **8B** are offset with respect to each other. The transmission mechanism **4** includes a link mechanism **33**. The second drive shaft **8B** is connected to the first drive shaft **8A** via the link mechanism **33** such as gears or a belt.

(32) In the outboard motor **1** according to the sixth preferred embodiment, the first pump **25** is connected to the second drive shaft **8B**. The second pump **26** is connected to the propeller shaft **9** as in the first preferred embodiment. Alternatively, the second pump **26** may be located similarly to the second to fifth preferred embodiments. FIG. **8** is a side view of the outboard motor **1** according to a seventh preferred embodiment. FIG. **9** is a side view of the outboard motor **1** according to an eighth preferred embodiment. FIG. **10** is a side view of the outboard motor **1** according to a ninth preferred embodiment. FIG. **11** is a side view of the outboard motor **1** according to a tenth preferred embodiment. The arrangement of the first pump **25** and the second pump **26** in the seventh to tenth preferred embodiments is the same as in the second to fifth preferred embodiments, respectively. As shown in FIGS. **8** to **11**, the second pump **26** may be connected to the first drive shaft **8A** or may be connected to the second drive shaft **8B**.

(33) In the first to tenth preferred embodiments described above, the first pump **25** may be a positive displacement pump and the second pump **26** may be a non-positive displacement pump. Alternatively, the first pump **25** may be a non-positive displacement pump and the second pump **26** may be a positive displacement pump. Alternatively, the first pump **25** and the second pump **26** may be non-positive displacement pumps. Alternatively, the first pump **25** and the second pump **26** may be positive displacement pumps.

(34) The configuration of the cooling water passage **16** is not limited to that of the above preferred embodiments, and may be modified. For example, the number of water intakes is not limited to one, and may be two or more. The first passage **22** and the second passage **23** may be connected to separate water intakes. The first passage **22** and the second passage **23** do not have to join downstream of the pumps **25** and **26**. That is, the first passage **22** and the second passage **23** may be connected to the water jacket **15** of the engine **3** independently of each other.

(35) The second pump **26** may send the cooling water to the first pump **25**. For example, if the first pump **25** is a non-positive displacement pump, the second pump **26** may deliver the cooling water to the first pump **25** as priming water to start the first pump **25**.

(36) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. An outboard motor comprising: an engine; an upper case below the engine; a lower case below the upper case; a cavitation plate connected to the lower case; a transmission including: a drive shaft extending downward from the engine and located in the upper case and the lower case; and a propeller shaft connected to the drive shaft, extending in a front-rear direction of the outboard motor, and located in the lower case; a cooling water passage connected to the engine and located in the upper case and the lower case; a first pump at least partially above the cavitation plate, connected to the drive shaft, drivable by a driving force from the drive shaft, and operable to send cooling water to the engine through the cooling water passage; and a second pump drivable by a

driving force from the transmission to send the cooling water to the engine through the cooling water passage.

2. The outboard motor according to claim 1, wherein the first pump is a positive displacement pump.
 3. The outboard motor according to claim 1, wherein the first pump is a non-positive displacement pump.
 4. The outboard motor according to claim 1, wherein the second pump is a positive displacement pump; and at least a portion of the second pump is above the cavitation plate.
 5. The outboard motor according to claim 4, wherein the engine includes a crankshaft; and the second pump is connected to the crankshaft.
 6. The outboard motor according to claim 4, wherein the transmission further includes an intermediate shaft offset from the drive shaft and connected to the drive shaft; and the second pump is connected to the intermediate shaft.
 7. The outboard motor according to claim 4, wherein the second pump is connected to the drive shaft.
 8. The outboard motor according to claim 1, further comprising: a bracket to mount the outboard motor to a watercraft; wherein the second pump is a positive displacement pump; and at least a portion of the second pump is below a lower end of the bracket.
 9. The outboard motor according to claim 8, wherein the second pump is connected to the drive shaft.
 10. The outboard motor according to claim 8, wherein the second pump is connected to the propeller shaft.
 11. The outboard motor according to claim 1, further comprising: a bracket to mount the outboard motor to a watercraft; wherein the second pump is a non-positive displacement pump; and at least a portion of the second pump is below a lower end of the bracket.
 12. The outboard motor according to claim 11, wherein the second pump is connected to the drive shaft.
 13. The outboard motor according to claim 11, wherein the second pump is connected to the propeller shaft.
 14. The outboard motor according to claim 1, wherein the first pump is coaxial with the drive shaft.
 15. The outboard motor according to claim 1, wherein the second pump is coaxial with the drive shaft.
 16. The outboard motor according to claim 1, wherein the second pump is coaxial with the propeller axis.
 17. The outboard motor according to claim 1, wherein the first pump is a non-positive displacement pump; the second pump is a positive displacement pump; at least a portion of the second pump is above the cavitation plate; the engine includes a crankshaft; and the second pump is connected to the crankshaft.
 18. The outboard motor according to claim 1, wherein the first pump is a non-positive displacement pump; the second pump is a positive displacement pump; at least a portion of the second pump is above the cavitation plate; the transmission further includes an intermediate shaft offset from the drive shaft and connected to the drive shaft; and the second pump is connected to the intermediate shaft.
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